

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B.Tech****Branch : Electronics & Computer Engineering/Electronics & Computer Science****Semester :V****Subject Code & Name: BTECOE504C PROGRAMMING IN JAVA****Max Marks: 60****Date: 13/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Who invented Java Programming?		1
	a. Guido van Rossum b. James Gosling c. Dennis Ritchie d. Bjarne Stroustrup		
2	Which component is used to compile, debug and execute the java programs?		1
	a. JRE b. JIT c. JDK d. JVM		
3	Which one of the following is not a Java feature?		1
	a. Object-oriented b. Use of pointers c. Portable d. Dynamic and Extensible		
4	What is the extension of java code files?		1
	a. .js b. .txt c. .class d. .java		
5	Which of the following is not an OOPS concept in Java?		1
	a. Polymorphism b. Inheritance c. Compilation d. Encapsulation		
6	Which exception is thrown when java is out of memory?		1
	a. Memory Error b. Out Of Memory Error c. Memory Out Of Bounds d. Memory Full Exception		

			Exception			
7	Which one of the following is not an access modifier?					1
	a. Protected	b. Void	c. Public	d. Private		
8	Which of these keywords is used to define interfaces in Java?					1
	a. intf	b. intfe	c. interface	d. intfer		
9	Which of these keywords is used to make a class?					1
	a. class	b. struct	c. int	d. none of the mentioned		
10	Which of the following is a valid declaration of an object of class Box?					1
	a. Box obj = new Box();	b. Box obj = new Box;	c. obj = new Box();	d. new Box obj;		
11	Which of these operators is used to allocate memory for an object?					1
	a. malloc	b. alloc	c. new	d. give		
12	Which of this keyword can be used in a subclass to call the constructor of superclass?					1
	a. super	b. this	c. extent	d. extends		
Q. 2	Solve the following.					12
A)	Explain in detail JVM,JDK and JRE in deatil					6
B)	Explain in detail data types in java					6
Q.3	Solve the following.					12
A)	Write a java program to count even and odd numbers					6
B)	Explain in detail operators in java					6
Q.4	Solve Any Two of the following.					12
A)	Explain class and object in detail with examples					6
B)	Explain in detail types of constructors					6
C)	Explain in detail types of variables in java					6

Q.5	Solve Any Two of the following.		12
A)	Explain in detail inheritance in java		6
B)	Explain in detail polymorphism in java		6
C)	Explain in detail multithreading		6
Q.6	Solve Any Two of the following.		12
A)	Explain in detail concept of exception handling in java		6
B)	Explain in detail concept of Applets		6
C)	Explain in detail JDBC driver		6
	*** End ***		